

Slide 1: Title & Introduction

Speaker Notes (Speaker 1):

"Hello everyone. Today, my partner and I will be talking about one of the most fascinating applications of particle-based simulations: Molecular Communication.

As computer engineers, we are used to devices communicating via electromagnetic waves. But what if we need to send data inside the human bloodstream using nano-robots? Radio waves don't work well there. Instead, these tiny machines communicate by releasing chemical molecules into the environment, much like sending messages using bursts of perfume.

Today, we will discuss the biggest problem in this field, how traditional math solved it, and how a modern Machine Learning approach completely changed the game. Finally, we'll explain how we plan to extend this for our final project."

Slide 2: The Core Challenge: Inter-Symbol Interference (ISI)

Speaker Notes (Speaker 1):

(Point to the slide / imaginary graph) "Let's understand the core problem. The transmitter uses a simple logic: it releases a burst of molecules for a bit '1', and releases nothing for a bit '0'. However, these molecules travel through random Brownian motion. They don't just disappear after the signal is received. They linger in the environment. This creates a 'heavy tail' in the signal. When you try to send the next bit, the old molecules mix with the new ones. We call this 'Inter-Symbol Interference', or ISI. If you want to send data faster—meaning a shorter time between bits—this chemical echo gets so loud that the receiver gets completely confused and baseline concentration drifts unpredictably."

Slide 3: Paper 1 (Meng 2014) - Static Channel Solutions

Speaker Notes (Speaker 1):

"Our first paper, by Meng and colleagues in 2014, offers a very elegant, mathematical solution to this ISI problem. They assume the environment is static. Since nano-devices have almost zero computational power, they can't run heavy algorithms. So, Meng proposed a technique called 'ISI Cancellation'. The logic is surprisingly simple: The receiver measures the molecule concentration at the very beginning of the time slot. This represents the 'noise' or the leftover molecules from previous signals. Then, it measures the concentration at the peak of the new signal. By simply subtracting the initial baseline from the peak, it cancels out the history and isolates the current bit. It's a brilliant, lightweight mathematical trick."

Slide 4: The Mobile Channel Problem

Speaker Notes (Speaker 2):

"Now, Meng's solution is mathematically beautiful, but it has one major flaw: it assumes the transmitter and receiver are just sitting still. In a real biological environment, like a blood vessel, nano-nodes are constantly moving due to Brownian motion. Because they are moving, the distance changes, and the channel's impulse response changes every single second. In such a dynamic environment, strict mathematical formulas and fixed thresholds completely collapse. You can't calculate a perfect threshold if the channel keeps shifting. To solve this, we must move from mathematical modeling to a data-driven approach: Machine Learning."

Slide 5: Paper 2 (Bilge 2025) - Data-Driven Detection

Speaker Notes (Speaker 2):

"This brings us to our second paper, published just recently in 2025 by Bilge, Akyol, and Kuscu. We know Machine Learning can learn dynamic environments. But here is the dilemma: you cannot put a complex LSTM or a Deep Convolutional Network inside a nanoscale robot. They require too much battery, memory, and computational power. The authors propose a brilliant alternative called 'Reservoir Computing'. It gives us the temporal memory of a Recurrent Neural Network, but it is incredibly cheap to compute and train. Let's see how it works."

Slide 6: Reservoir Computing (RC) Architecture

Speaker Notes (Speaker 2 - Samed):

(Explain this part slowly, as it's the core of the project) "Imagine the 'Reservoir' as a pool of water. When new molecules arrive, it's like throwing a stone into the pool. The ripples mix with the older ripples. In Reservoir Computing, we have a network of nodes, but we *do not* train their weights. They are fixed randomly. They just act as a physical memory that naturally tracks how the concentration changes over time. All we do is train one single linear layer at the very end—the readout layer—to look at the state of the pool and decide if it's a '1' or a '0'. Because we skip heavy backpropagation, the training cost is almost zero."

Slide 7: Simulation Environment (Smoldyn)

Speaker Notes (Speaker 1):

"To test this, the authors used a framework that perfectly aligns with our course: Smoldyn, which is a 3D particle-based stochastic simulator. They simulated a realistic 3D space where the transmitter and receiver are moving randomly. For every bit '1', 2000 molecules are released. To test the system to its limits, they changed the symbol duration—meaning the time between bits. They tested it from 200 seconds, where it's easy to read, all the way down to 10 seconds, where the molecules are so densely packed that the ISI noise is extreme."

Slide 8: Superior Results & Efficiency

Speaker Notes (Speaker 1):

"The results of the simulation are highly impressive. Let's look at the worst-case scenario: the 10-second interval where ISI is extremely severe. Traditional threshold methods and even deep learning models like LSTM drop to around 50% accuracy—which, in binary, means they are just randomly guessing. But the RC model maintained its robustness. What's more impressive is the cost. A standard Multilayer Perceptron (MLP) needed up to 264,000 parameters to run this. The Reservoir Computing model? Only 401 parameters. It processes a symbol in just 1 microsecond. It is fast, lightweight, and highly accurate."

Slide 9: Our Final Project Proposal (Task 3 Extension)

Speaker Notes (Speaker 2):

"So, how are we going to extend this for our final project?"

First, we will implement this particle-based simulation environment in Python and build the base Reservoir Computing model.

For our 'novel extension', we plan to create a Hybrid Architecture. We will take the mathematical 'ISI Cancellation' technique from the 2014 paper, use it as a pre-processing filter to clean the obvious noise, and then feed that 'cleaned' data into the Reservoir Computer from the 2025 paper.

We believe this combination of classical physics-based math and modern Machine Learning will increase the accuracy even further. We will implement both and present the comparison metrics as our final project."

Slide 10: Conclusion & Q&A

Speaker Notes (Both):

(Speaker 1): "To summarize, sending messages with molecules is messy, but by using smart, low-cost AI like Reservoir Computing, we can make nano-devices communicate reliably even while moving in the bloodstream."

(Speaker 2): "This concludes our presentation. Thank you for listening. If you have any questions or feedback about our extension idea, we'd love to hear them."